5. THE NUCLEAR FORCES

I. GENERAL PROPERTIES OF THE NUCLEAR FIELD

The cohesion of protons and neutrons within the atomic nucleus is ensured by forces, to which we have referred as *nuclear forces,* although we have not yet discussed their nature. The electric forces of repulsion, which are also operative in the nucleus, exert a purely disruptive effect. What can we learn today from the experiments on the nature of these nuclear forces? Let us first study briefly the form which an answer to this question must take. Assuming that we did not as yet know what electric forces were, how could we begin to explain their nature? We could begin by stating that electric charges mutually repel each other, with a force which decreases inversely as the square of the distance between them. According to the knowledge gained in the early part of the nineteenth century, we could add that a fundamental relationship exists between electric and magnetic forces—e.g.: varying electric forces always generate magnetic forces, and vice versa. Moreover, we might add that the phenomena of light, which used to be regarded as something of a special nature, are among these electromagnetic phenomena, and are simply nothing but electromagnetic waves. The next step would be to note that in certain experiments light appears not as a wave, but as flying particles, in other words, in the shape of photons, and thus we would discover a relationship between the electromagnetic field and these photons. A truly exhaustive description of electromagnetic forces can, however, be given solely with the aid of mathematical equations, expressing how electromagnetic forces change and spread. A complete picture of the ‘nature’ of these phenomena is supplied only by Maxwell’s equations in combination with the equations of the quantum theory.

But as for the nuclear forces we have not got to them yet. Nevertheless, we can already form a picture of them, which is correct qualitatively at least and includes as many details as the picture of the electromagnetic forces—except for the exact mathematical formulae.

The first question to arise here is: How does the force operating between two particles in the atomic nucleus depend on their distance from each other? Is it perhaps also in inverse proportion to the square of this distance? The simplest object available for the study of this problem is the deuteron; we inquire about the force which binds a proton and a neutron in a deuteron. Once this force is known, we have a good prospect of comprehending the cohesion of other nuclei. The force in question cannot possibly be electrical in